

LAB EXPERIMENT ITA0506-COMPUTER VISION

P.POORNA CHANDRA-192472287

EXPNO: 32. Morphological Operation Using Closing

Aim

To perform morphological closing on an image using OpenCV.

Algorithm

1. Import OpenCV.
2. Read BEN 10.png.
3. Convert the image to grayscale.
4. Create a morphological kernel.
5. Apply closing using cv2.morphologyEx().
6. Display and save the result.

CODE:



```
EXP-32.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...
File Edit Format Run Options Window Help
import cv2
import numpy as np

# Read image
img = cv2.imread("BEN 10.png")

if img is None:
    print("Image not found!")
    exit()

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Create kernel
kernel = np.ones((5, 5), np.uint8)

# Apply closing
closing = cv2.morphologyEx(gray, cv2.MORPH_CLOSE, kernel)

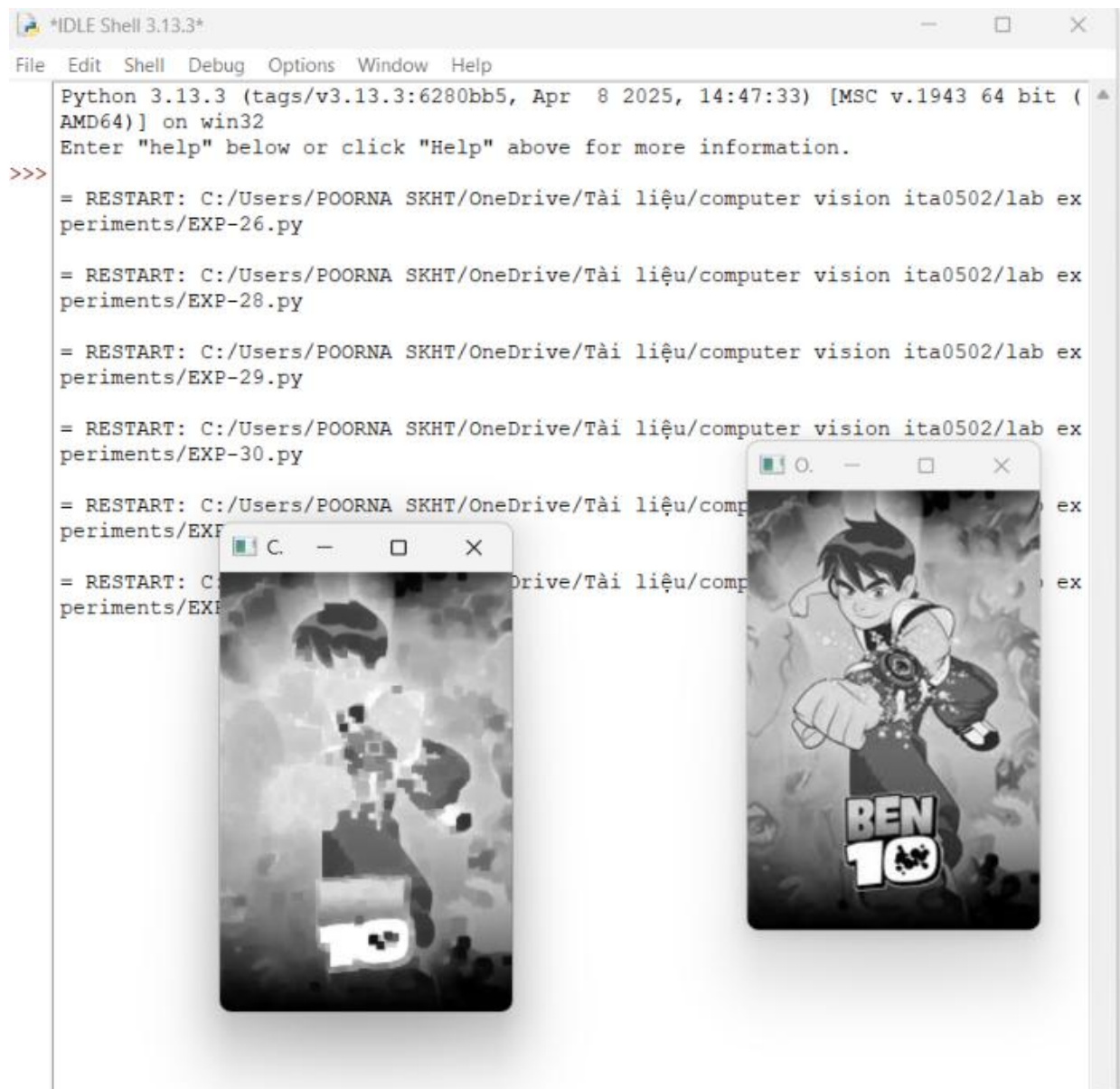
# Display result
cv2.imshow("Original Image", gray)
cv2.imshow("Closing", closing)

# Save result
cv2.imwrite("closing.png", closing)

cv2.waitKey(0)
cv2.destroyAllWindows()

print("Closing completed successfully!")
```

OUTPUT :



The screenshot shows an IDLE Shell window with the following content:

```
*IDLE Shell 3.13.3*
File Edit Shell Debug Options Window Help
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr 8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-26.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-28.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-29.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-30.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-31.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-32.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-33.py
```

Two small windows are overlaid on the shell, each displaying a grayscale image of Ben 10. The left window shows a noisy, pixelated version of the character, while the right window shows a cleaner, more defined version of the same character, demonstrating the result of a morphological opening operation.

Result

Morphological opening is successfully performed. It helps remove small unwanted noise from the image.